Experiment-3.2

Aim: - To Study and Perform basic pen testing

Mapped CO: - Understand the different types of Attacks, Vulnerabilities. **CO3**

Objective

- To understand the concept of penetration testing and its importance in cyber security.
- To study the basic phases of penetration testing.
- To perform basic penetration testing using commonly used security tools.
- To identify vulnerabilities in a target system in a controlled and ethical manner

Theory

Penetration Testing

Penetration Testing (Pen Testing) is a **systematic process of evaluating the security of a computer system, network, or web application by simulating real-world attacks**. The objective is to identify vulnerabilities before malicious attackers can exploit them.

Penetration testing helps organizations:

- Identify security weaknesses
- Validate security controls
- Improve risk management
- Ensure compliance with security standards

Types of Penetration Testing

1. **Black Box Testing** – Tester has no prior knowledge of the system.
2. **White Box Testing** – Tester has complete knowledge of the system.
3. **Gray Box Testing** – Tester has partial knowledge of the system.

4. Phases of Penetration Testing

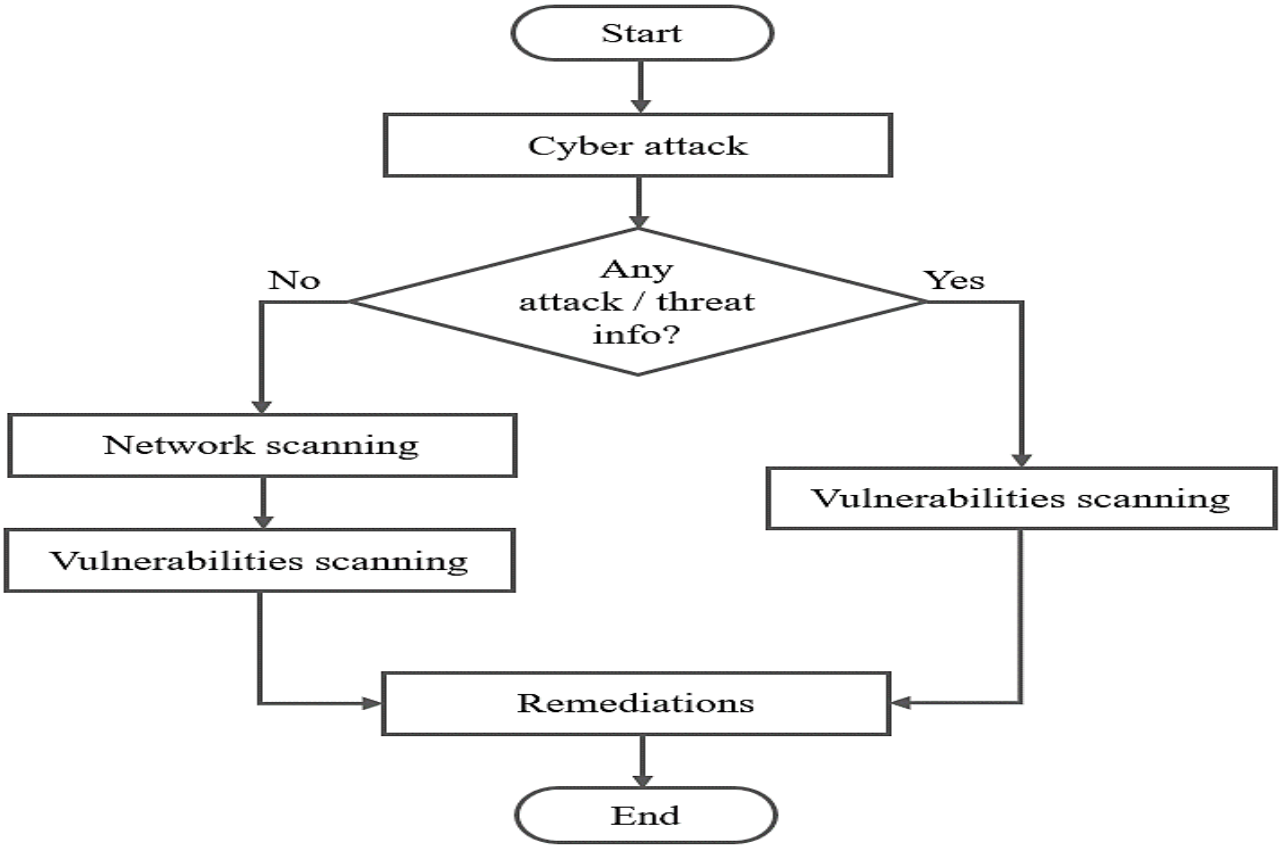
Phase 1: Reconnaissance (Information Gathering)

- Collect information about target.
- Passive: Google search, WHOIS lookup.
- Active: Network probing.

Tools:

- whois
- nslookup
- theHarvester

Phase 2: Scanning



- Identify open ports and services.
- Detect operating systems and vulnerabilities.

Tools:

- Nmap
- Netcat

Example:

```
nmap -sV <target IP>
```

Phase 3: Vulnerability Analysis

- Analyze discovered services.
- Find security weaknesses.

Tools:

- OpenVAS
- Nikto
- Nessus

Phase 4: Exploitation (Basic Level – Demonstration Only)

- Attempt controlled exploitation.
- Validate vulnerabilities.

Tool:

- Metasploit Framework

Phase 5: Reporting

- Document findings.
- Provide remediation steps.

Report Includes:

- Vulnerabilities found
- Risk level
- Evidence
- Recommendations

Step 6: Documentation

Record:

- Commands used
- Screenshots
- Vulnerabilities found
- Risk rating

5. Result

Open ports, services, and vulnerabilities were identified and documented.

6. Precautions

- Perform testing only with permission.
- Use isolated virtual lab.
- Do not attack real systems.
- Follow ethical hacking guidelines.

7. Applications

- Security auditing
- Vulnerability assessment
- Network hardening
- Compliance testing

Result

Open ports, running services, and potential vulnerabilities of the target system were identified using Nmap and Nikto. This experiment demonstrated how penetration testing helps in improving system security by identifying weaknesses before attackers exploit them.

YouTube Video Links (Reference)

1. Basic Penetration Testing Tutorial (Nmap & Kali Linux)
   https://www.youtube.com/watch?v=4t4kBkMsDbQ
2. Introduction to Penetration Testing
   https://www.youtube.com/watch?v=3Kq1MIfTWCE

(For educational purposes only)

Viva Questions

1. What is penetration testing?
2. Why is penetration testing important for organizations?
3. What are the different types of penetration testing?
4. What is Nmap and why is it used?
5. What is vulnerability scanning?